

Permissions

Adapted from materials by Dr. Carrier



Permissions

Who can access your files?

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We break things into categories:

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- Others (everyone!)

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9

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Viewing permissions

How do we see what files we have?

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ls
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```
ls
```

ls has an extra flag to tell us more about files:

```
ls -l
```

```
-rw-rw-r--  1 ferguaus ferguaus      118 Sep  6 10:06 test_2.cpp
```

Breaking it down

```
-rw-rw-r-- 1 ferguaus ferguaus 118 Sep  6 10:06 test_2.cpp
```

```
-rw-rw-r-- 1
```

Breaking it down

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-rw-rw-r-- 1 ferguaus ferguaus 118 Sep 6 10:06 test_2.cpp
```

```
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```



What type of
file is this?

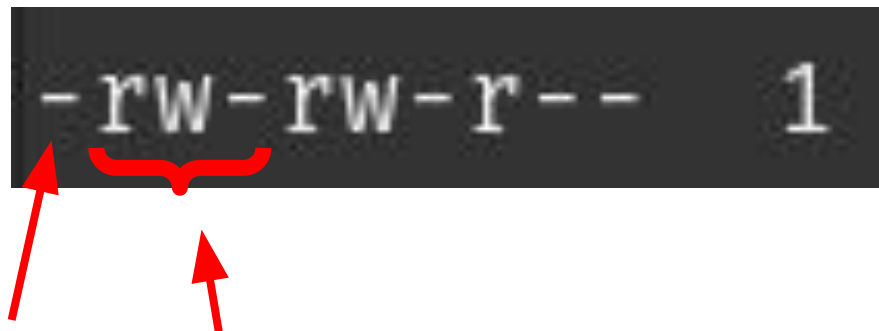
d = directory

l = link

Breaking it down

```
-rw-rw-r-- 1 ferguaus ferguaus 118 Sep 6 10:06 test_2.cpp
```

```
-rw-rw-r-- 1
```



What type of
file is this?

d = directory

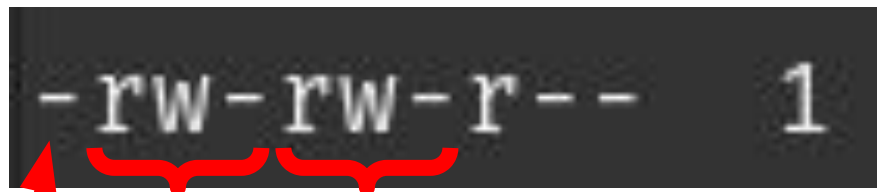
l = link

Owner
permissions

Breaking it down

```
-rw-rw-r-- 1 ferguaus ferguaus 118 Sep 6 10:06 test_2.cpp
```

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-rw-rw-r-- 1
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What type of
file is this?

d = directory

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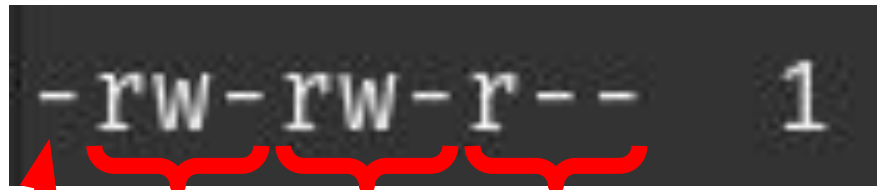
Owner
permissions

Group
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Breaking it down

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-rw-rw-r-- 1 ferguaus ferguaus 118 Sep 6 10:06 test_2.cpp
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What type of
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d = directory

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Owner
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Group
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Other
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Breaking it down

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Number of links

Owner
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ferguaus ferguaus
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Breaking it down

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-rw-rw-r-- 1 ferguaus ferguaus 118 Sep 6 10:06 test_2.cpp
```

`-rw-rw-r-- 1`

Number of links

Owner permissions Group permissions Other permissions

What type of file is this?
d = directory
l = link

`ferguaus ferguaus`

`118 Sep 6 10:06 test_2.cpp`

Owner Group File size

Breaking it down

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-rw-rw-r-- 1 ferguaus ferguaus 118 Sep 6 10:06 test_2.cpp
```

```
-rw-rw-r-- 1
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Number of links

Owner permissions Group permissions Other permissions

What type of file is this?
d = directory
l = link

```
ferguaus ferguaus
```

```
118 Sep 6 10:06 test_2.cpp
```

Owner Group File size Modified date+time

Breaking it down

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-rw-rw-r-- 1 ferguaus ferguaus 118 Sep 6 10:06 test_2.cpp
```

`-rw-rw-r-- 1`

Number of links

Owner permissions Group permissions Other permissions

What type of file is this?
d = directory
l = link

`ferguaus ferguaus`

Owner Group

`118 Sep 6 10:06 test_2.cpp`

File size Modified date+time Filename

Examples

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`rxwxrx-rx`

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`rxwxrx-r-x`

`rw-r-----`

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`rxwxrwxr-x`

`rw-r-----`

`-----rw-`

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Who: owner (u), group (g), other (o), all (a)

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Who: owner (u), group (g), other (o), all (a)

Perms: read (r), write (w), execute (x)

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Usage 1: chmod (who)(op)(perms) file

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chmod g+r file
```

```
chmod o-rw file
```

Who: owner (u), group (g), other (o), all (a)

Perms: read (r), write (w), execute (x)

Give a command to do each for test.sh

- Allow everyone to write
- Take away execute permission for the group
- Make sure that only the user can read the file

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Usage 2: `chmod ### file`

Where each # is a number in $[0, 7]$

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Where each # is a number in $[0, 7]$

Take each possible permission:

- 4 = read
- 2 = write
- 1 = execute

And sum them up for owner, group, other

Examples

What is the number code for each of these?

`rw-rwxr-x` `775`

`rw-r-----`

`-----rw-`

- 4 = read
- 2 = write
- 1 = execute

Examples

What is the number code for each of these?

`rxwxr-x` `775`

`rw-r-----` `640`

`-----rw-`

- 4 = read
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Examples

What is the number code for each of these?

`rxwxr-x` 775

`rw-r-----` 640

`-----rw-` 006

- 4 = read
- 2 = write
- 1 = execute

Examples

What is the number code for each of these?

`rw-rwxr-x` `775`

`rw-r-----` `640`

`-----rw-` `006`

- 4 = read
- 2 = write
- 1 = execute

Go the other way:

What are the permissions for `541`

More practice!

What does each command do?

- `chmod u+x file.sh`
- `chmod og-rw file.sh`
- `chmod a=r file.sh`
- `chmod 520 file.sh`

Give the command to:

- Set permissions of `test.sh` to `rwxr--r-x`
- Remove write permission on `test.sh` from other